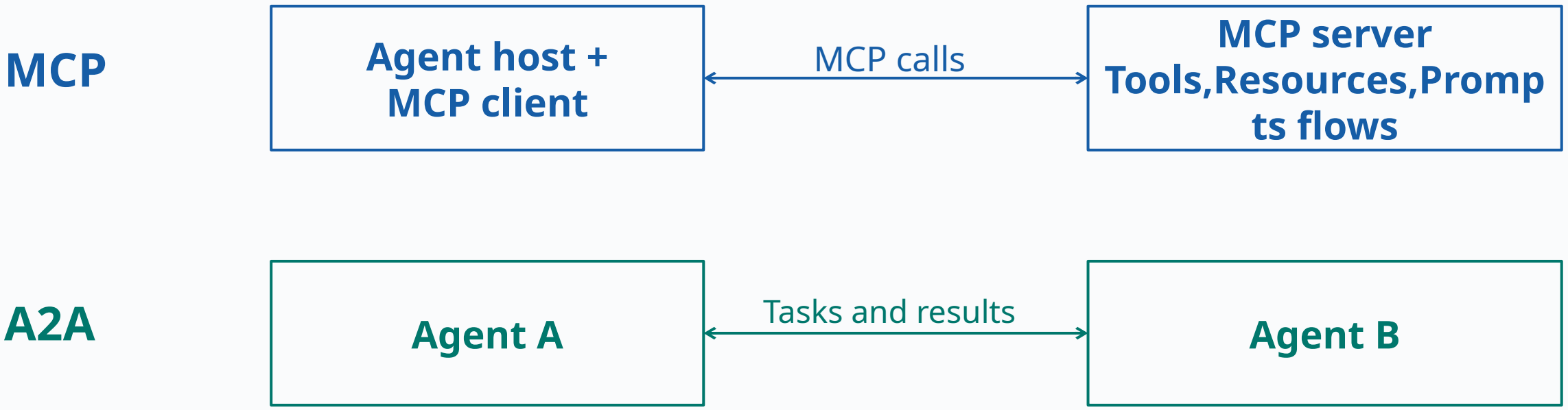
MCP and A2A: the big picture

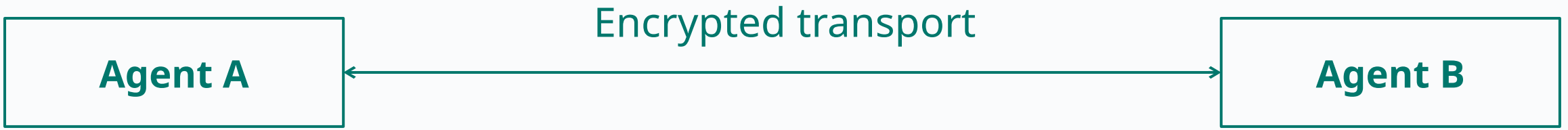
Two application protocols, with encryption provided by the transport



Remote HTTPS deployment: HTTP messages travel inside TLS.

A2A requires encryption in production

HTTPS for HTTP bindings
TLS for gRPC



TLS 1.3+ is recommended.

A gateway that terminates TLS can read the messages.

Source: A2A specification, §§7.1–7.2. Requirement ≠ verification of every deployment.

MCP capture: encrypted traffic to Canva

Real Codex MCP client and Canva server



Wireshark

Encrypted application data
Visible connection metadata

Authorized proxy

Readable HTTP and MCP JSON

The tested client (CODEX – OpenAI) trusted the temporary proxy CA. No TLS downgrade.

Inside a real MCP exchange

Old protocol version: 2025-06-18. HTTP/2.

initialize / notifications/initialized / tools/list / tools/call

Request

```
{
  "jsonrpc": "2.0",
  "id": 2,
  "method": "tools/call",
  "params": {
    "name": "search-designs",
    "arguments": {"limit": 1}
  }
}
```

Response

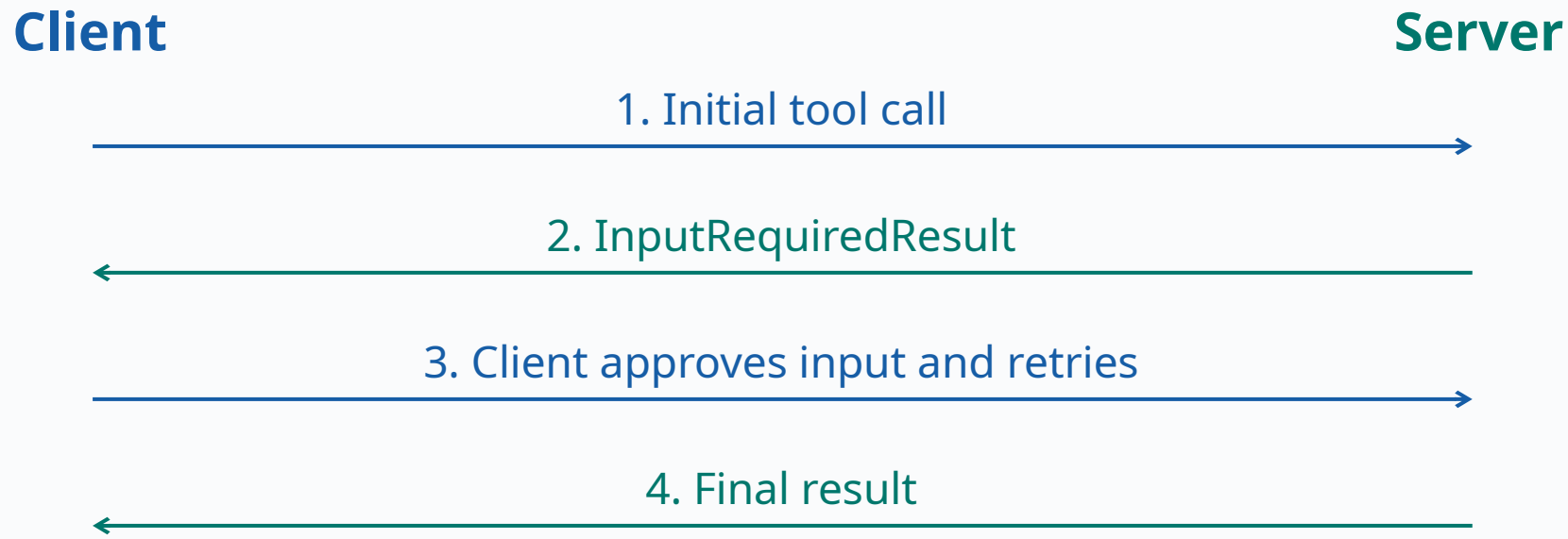
```
{
  "jsonrpc": "2.0",
  "id": 2,
  "result": {
    "content": [{
      "type": "text",
      "text": "[REDACTED]"
    }]
  }
}
```

MRTR in MCP 2026-07-28

The July release replaces server-initiated requests with explicit input rounds.

No more initialize handshake or MCP session ID

Each request carries its version and capabilities.



TLS remains a separate layer. SSE responses still exist (for notifications).

MRTR in JSON

Illustrative exchange for a workspace tool.

1 Initial call

```
{"id": 1, "method": "tools/call",
 "params": {"name": "inspect"}}
```

2 Server asks for roots

```
{
 "id": 1,
 "result": {
  "resultType": "input_required",
  "inputRequests": {
   "r": {"method": "roots/list"}
  },
  "requestState": "Js255d22d552f2"
 }
}
```

3 Client retries with a new id

```
{
 "id": 2,
 "params": {
  "name": "inspect",
  "inputResponses": {
   "r": {"roots": []}
  },
  "requestState": "Js255d22d552f2"
 }
}
```

How many requests does MCP need?

Published task-level data exists. A universal network-request average remains unestablished.

17.4

tool calls per task

MCPMark, 2025
127 benchmark tasks

The unit matters

Tool calls /= HTTP requests

HTTP requests /= TCP packets

Task, model and protocol version all affect the count.

Source: Wu et al., MCPMark, arXiv:2509.24002. Benchmark average.

TLS evidence inside the capture

Packet-level findings from the supplied Canva capture

ClientHello

Frames 8 and 12
Server name: mcp.canva.com

ServerHello

Frames 15 and 19
Selected version: TLS 1.3

The capture exposes handshake metadata.
The HTTP headers and MCP bodies remain encrypted.

```
Wireshark filter: tls.handshake.type == 2
```

Source: canva_only.pcapng

Captured initialization: request and response (OLD)

MCP 2025-06-18. The response repeats the request identifier.

Client request


```
{
 "jsonrpc": "2.0",
 "id": 0,
 "method": "initialize",
 "params": {
  "protocolVersion": "2025-06-18",
  "capabilities": {
   "elicitation": {"form": {}, "url": {}}
  },
  "clientInfo": {
   "name": "codex-mcp-client",
   "title": "Codex",
   "version": "0.154.0-alpha.6.2"
  }
 }
}
```


Server response


```
{
 "jsonrpc": "2.0",
 "id": 0,
 "result": {
  "protocolVersion": "2025-06-18",
  "capabilities": {
   "tools": {"listChanged": false},
   "resources": {"listChanged": true}
  },
  "serverInfo": {
   "name": "Canva MCP Server",
   "version": "1.0.0"
  }
 }
}
```


Source: mcp_initialize_request.json

Captured tool call: search-designs (OLD)

Request id 2. Metadata and tool arguments occupy different fields.

Request envelope


```
{
 "jsonrpc": "2.0",
 "id": 2,
 "method": "tools/call",
 "params": {
  "name": "search-designs",
  "_meta": {"progressToken": 1}
 }
}
```


_meta carries request metadata.
arguments carries inputs for the tool.

params.arguments


```
{
 "query":
  "MCP_CAPTURE_TEST_20260929_DIRECT_91cd",
 "sort_by": "relevance",
 "ownership": "any",
 "limit": 1
}
```


Source: mcp_tool_request.json

MRTR example: the initial tool call (NEW)

Illustrative MCP 2026-07-28 exchange, independent of the Canva capture

Client request, id 1

```
{
 "jsonrpc": "2.0",
 "id": 1,
 "method": "tools/call",
 "params": {
  "name": "create_report",
  "arguments": {
   "topic": "MCP and A2A security",
   "format": "pdf"
  }
 }
}
```

params._meta

```
{
 "io.modelcontextprotocol/protocolVersion":
  "2026-07-28",
 "io.modelcontextprotocol/clientInfo": {
  "name": "mrtr-demo-client",
  "version": "1.0.0"
 },
 "io.modelcontextprotocol/clientCapabilities": {
  "elicitation": {"form": {}}
 }
}
```

The client advertises support for form elicitation.

Source: mrtr_communication_example.json

MRTR example: the server requests input (NEW)

InputRequiredResult completes the first request and asks for a confirmation.

Response, id 1

```
{
 "jsonrpc": "2.0",
 "id": 1,
 "result": {
  "resultType": "input_required",
  "requestState": "DEMO_OPAQUE_STATE"
 }
}
```

inputRequests.confirm_report

```
{
 "method": "elicitation/create",
 "params": {
  "mode": "form",
  "requestedSchema": {
   "type": "object",
   "properties": {
    "confirmed": {"type": "boolean"}
   },
   "required": ["confirmed"]
  }
 }
}
```

The form asks whether to create the PDF report.
The client decides whether to collect input and retry.

Source: mrtr_communication_example.json

MRTR example: retry and final result (NEW)

The second request uses id 2 and returns the same opaque state.

Client retry

```
{
 "jsonrpc": "2.0",
 "id": 2,
 "method": "tools/call",
 "params": {
  "name": "create_report",
  "inputResponses": {
   "confirm_report": {
    "action": "accept",
    "content": {"confirmed": true}
   }
  },
  "requestState": "DEMO_OPAQUE_STATE"
 }
}
```

Server result

```
{
 "jsonrpc": "2.0",
 "id": 2,
 "result": {
  "resultType": "complete",
  "isError": false,
  "structuredContent": {
   "reportId": "demo-report-001",
   "format": "pdf",
   "status": "created"
  }
 }
}
```

Source: mrtr_communication_example.json